

CLASS 1

API Basics Before Django REST Framework

Understand the purpose first. Code comes later.

STARTING POINT

What is Django for?

Django mainly builds the **backend** of a web application.

It handles

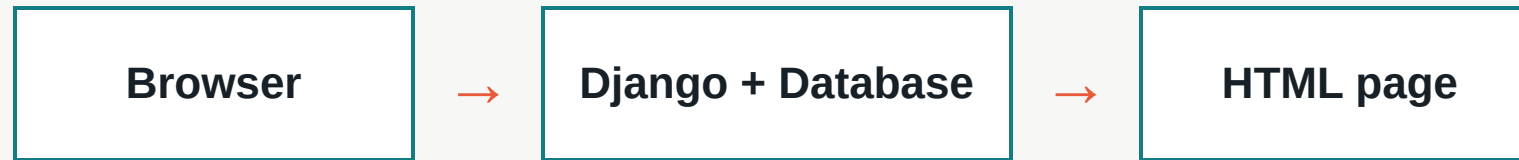
URLs, logic, users, and database work.

It can also

Create HTML pages using templates.

TRADITIONAL DJANGO

Normal Django returns a ready HTML page



The template decides how the database data looks to a human.

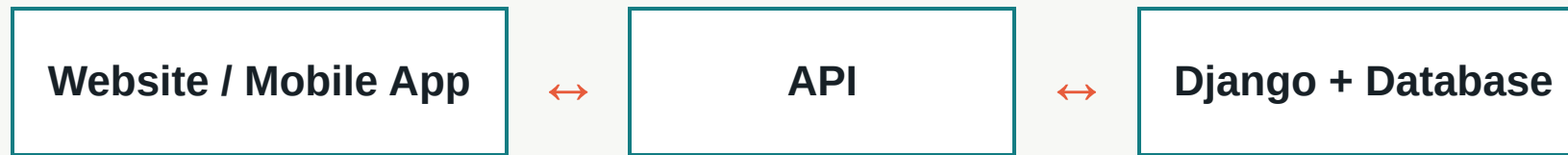
```
/account/customer/ -> ready HTML page
```

API

What is an API?

Application Programming Interface

An API is a way for one software system to talk to another software system.



JSON

API usually returns data, not page design

HTML

Ready page for a human.

JSON

Structured data for software.

```
{  
  "id": 5,  
  "name": "Student"  
}
```

MAIN PURPOSE

Why use an API?

The main reason is **separation and reuse**, not security or speed.

Website

Uses the JSON.

Mobile app

Uses the same JSON.

Other software

Uses the same JSON.

Each client decides how to display the data.

HTTP

How do clients ask an API?

They send an **HTTP request**. The method says what they want to do.

Method	Meaning	CRUD
GET	Read data	Read
POST	Create data	Create
PUT / PATCH	Update data	Update
DELETE	Delete data	Delete

REST

What is a REST API?

A REST API uses URLs and HTTP methods in a clear, predictable way.

```
GET    /registrations/    -> read all
POST   /registrations/    -> create
GET    /registrations/5/  -> read one
PATCH /registrations/5/   -> update part
DELETE /registrations/5/   -> delete
```

TOOLS

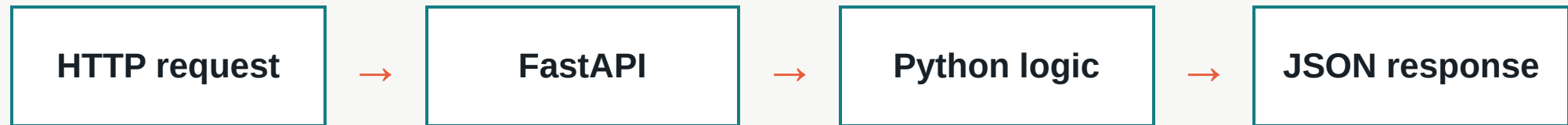
Django, DRF, and FastAPI

Tool	Simple meaning
Django	Full Python web framework; can return HTML.
DRF	Django package that makes REST APIs easier.
FastAPI	Separate Python framework focused on APIs.

DRF and FastAPI both build APIs. We use DRF because this project is already Django.

What is FastAPI?

FastAPI is a Python framework mainly made for building APIs.



It validates incoming data and can generate API documentation automatically.

FASTAPI EXAMPLE

A very small FastAPI endpoint

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/hello")
def hello():
    return {"message": "Hello"}
```

GET /hello returns JSON: {"message": "Hello"}

FASTAPI FEATURES

What does FastAPI give us?

Routes

Connect methods and URLs to Python functions.

Validation

Checks incoming request data.

API docs

Provides an interactive /docs page.

FastAPI is the tool. REST is the API design style.

REQUEST

Method + URL is one API instruction

```
GET /account/api/registrations/
```

Meaning: “Give me all registrations.”

```
curl -X GET http://127.0.0.1:8001/account/api/registrations/
```

curl sends the request. -X GET chooses the method.

FINISH HERE

Connect today's lesson to our project

FastAPI

A separate Python framework for building APIs.

DRF

The API toolkit used in our existing Django CRM.

Both can build REST APIs that use HTTP methods and return JSON.

```
http://127.0.0.1:8001/account/api/registrations/
```

Next class: Serializer + ViewSet + Router in this project.